Distributional Reinforcement Learning | Part 2

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Outline

1. Recap
2. Scalar Linear Function Approximation
3. Deep Distributional Reinforcement Learning

Recap: MDP and Return

We are working on a finite MDP:

$$(\mathcal{X}, \mathcal{A}, \xi_0, P_X, P_R), \qquad \pi : \mathcal{X} \to \mathcal{P}(\mathcal{A}), \qquad \gamma \in [0, 1).$$

Trajectory process:

$$X_0 \sim \xi_0, \qquad A_t \sim \pi(\cdot \mid X_t),$$

$$R_t \sim P_R(\cdot \mid X_t, A_t), \qquad X_{t+1} \sim P_X(\cdot \mid X_t, A_t), \qquad t \geq 0.$$

Discounted return:

$$G^\pi(x) = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{given } X_0 = x.$$

Recap: Value function and Bellman Operator

The value function is given by the expected return:

$$V^\pi(x) = \mathbb{E}[G^\pi(x)] = \int_{\mathbb{R}} z \, d\eta_\pi(x)(z)$$

Where $\eta_\pi(x) = \mathcal{D}(G^\pi(x))$ is the return distribution.
As we know, there is a recursive relation for the value function known as Bellman's Equation:

$$V^\pi = \mathcal{T}^\pi V^\pi.$$

Where we have the Bellman operator $\mathcal{T}^\pi : \mathbb{R}^{\mathcal{X}} \to \mathbb{R}^{\mathcal{X}}$:

$$(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi \left[R + \gamma V(X') \mid X = x \right].$$

Recap: Random-Variable Bellman Equation

With (A, R, X') sampled from $\pi(\cdot \mid x) P_R(\cdot \mid x, A) P_X(\cdot \mid x, A)$, the random-variable Bellman equation is:

$$G^\pi(x) \stackrel{D}{=} R + \gamma G^\pi(X') \mid X = x.$$

Taking laws gives the return-distribution equation:

$$\eta_\pi(x) = \mathcal{D}\left(R + \gamma G^\pi(X') \mid X = x\right).$$

This is the distributional version of $(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi[R + \gamma V(X') \mid X = x]$.

Recap: Distributional Bellman Operator

Define the bootstrap map

$$b_{r,\gamma}(z) = r + \gamma z, \qquad r, z \in \mathbb{R},$$

and for $Z' \sim \eta(x')$,

$$(b_{r,\gamma})_{\#}\eta(x') = \mathcal{D}(r + \gamma Z').$$

Distributional Bellman operator $\mathcal{T}^\pi : \mathcal{P}(\mathbb{R})^{\mathcal{X}} \to \mathcal{P}(\mathbb{R})^{\mathcal{X}}$:

$$(\mathcal{T}^\pi \eta)(x) = \mathbb{E}_\pi \left[(b_{R,\gamma})_{\#}\eta(X') \mid X = x \right].$$

Distributional Bellman equation:

$$\eta_\pi(x) = \mathbb{E}_\pi \left[(b_{R,\gamma})_{\#}\eta_\pi(X') \mid X = x \right], \qquad \eta_\pi = \mathcal{T}^\pi \eta_\pi.$$

Recap: Representations of Return Distributions

Categorical representation:

$$\eta(x) = \sum_{i=1}^{m} p_i^\theta(x)\delta_{\theta_i}, \qquad p_i^\theta(x) \geq 0, \quad \sum_{i=1}^{m} p_i^\theta(x) = 1.$$

Quantile representation:

$$\eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}, \qquad \tau_i = \frac{2i-1}{2m}, \qquad \theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i).$$

Focus of this Presentation

If $\mathcal{X}$ is large, then storing even a scaler, let alone a distribution per state is expensive. Hence, we look for suitable approximations.

Linear Value Function Approximation | Scalar Case

For weights $w \in \mathbb{R}^n$ approximate the value function as

$$V_w(x) = \phi(x)^\top w, \qquad \nabla_w V_w(x) = \phi(x).$$

For finite $\mathcal{X}$ and feature map ϕ, stack feature rows into $\Phi \in \mathbb{R}^{|\mathcal{X}| \times n}$:

$$V_w = \Phi w, \qquad \mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}.$$

For state-action pairs, use e.g. $\phi : \mathcal{X} \times \mathcal{A} \to \mathbb{R}^n$ via $Q_w(x, a) = \phi(x, a)^\top w$.

Idea

Instead of having a lookup table for the value of each state, we use a linear function to represent the function. This function can be queried cheaply.

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Idea

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Question: How do we get a good w?

Linear Approximation as Regression

Fitting a linear value function is just ordinary least squares:

$$V_w = \Phi w, \qquad \min_{w \in \mathbb{R}^n} \| V^\pi - V_w \|_2.$$

which we could solve by the normal equation.

However

In reinforcement learning, we might want to have a better approximation for some states than others.

Weighted Norm for Linear Approximation

For some $\xi \in \mathcal{P}(\mathcal{X})$ we define the weighted L_2 norm and diagonal weight matrix

$$\|V - V'\|_{\xi,2} = \left(\sum_{x \in \mathcal{X}} \xi(x)(V(x) - V'(x))^2 \right)^{1/2}. \qquad \Xi = \text{diag}(\xi(x) : x \in \mathcal{X}).$$

The problem we now want to solve is:

$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi,2}.$$

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$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi,2}.$$

Proposition 9.3 | Weighted Normal Equation

If ξ has full support and the columns of Φ are linearly independent, the unique solution is

$$w^* = (\Phi^\top \Xi \Phi)^{-1} \Phi^\top \Xi V^\pi.$$

Projection Back into the Feature Space

We have a linear approximation now, however we cant iterate the Bellman operator because we could have

$$\mathcal{T}^\pi V_w \notin \mathcal{F}_\phi$$

.

Define the weighted orthogonal projection

$$(\Pi_{\phi,\xi} V)(x) = \phi(x)^\top w^*, \qquad w^* \in \arg\min_w \| V - V_w \|_{\xi,2}.$$

Approximate dynamic programming therefore iterates

$$V_{k+1} = \Pi_{\phi,\xi} \mathcal{T}^\pi V_k.$$

- $\mathcal{T}^\pi$ performs the Bellman update.
- $\Pi_{\phi,\xi}$ removes components that the features cannot express.

Repeated projection can compound error (the diffusion effect).

When Is the Projected Operator Stable?

Let ξ_π be a fully supported steady-state distribution under π.

Projected Bellman contraction / Theorem 9.8

Since P^π and Π_{ϕ,ξ_π} are nonexpansions in $\|\cdot\|_{\xi_\pi,2}$,

$$\|\Pi_{\phi,\xi_\pi}\mathcal{T}^\pi V - \Pi_{\phi,\xi_\pi}\mathcal{T}^\pi V'\|_{\xi_\pi,2} \leq \gamma\|V - V'\|_{\xi_\pi,2}.$$

Therefore there is a unique fixed point $\widehat{V}^\pi = \Pi_{\phi,\xi_\pi}\mathcal{T}^\pi\widehat{V}^\pi$, with

$$\|\widehat{V}^\pi - V^\pi\|_{\xi_\pi,2} \leq \frac{1}{\sqrt{1-\gamma^2}}\|\Pi_{\phi,\xi_\pi}V^\pi - V^\pi\|_{\xi_\pi,2}.$$

Scalar Linear Approximation: Proof Chain

The scalar argument works because everything lives in a Hilbert space:

$$\mathbb{R}^{\mathcal{X}}, \qquad \langle V, V' \rangle_{\xi_\pi} = \sum_{x \in \mathcal{X}} \xi_\pi(x) V(x) V'(x).$$

The feature class $\mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}$ is a linear subspace.

1. Π_{ϕ,ξ_π} is an orthogonal projection, hence nonexpansive.
2. Stationarity of ξ_π makes P^π nonexpansive in $\|\cdot\|_{\xi_\pi,2}$.
3. The Bellman operator is affine with linear part γP^π.

Therefore

$$\Pi_{\phi,\xi_\pi} \mathcal{T}^\pi \quad \text{is a } \gamma\text{-contraction.}$$

Temporal-Difference Learning: Sampled Bellman Updates

Computing the Bellman iterates is expensive because it requires the expectation over the next state and reward. Instead, TD learning uses the sampled one-step Bellman target

$$r_k + \gamma V_w(x'_k).$$

The TD error is target minus current prediction:

$$\delta_k = r_k + \gamma V_w(x'_k) - V_w(x_k).$$

Equivalently, TD minimizes the squared TD error

$$L_k(w) = \frac{1}{2}\delta_k^2 = \frac{1}{2}\left(r_k + \gamma V_w(x'_k) - V_w(x_k)\right)^2.$$

In the semi-gradient step, only $V_w(x_k)$ is differentiated:

$$w \leftarrow w + \alpha_k \delta_k \nabla_w V_w(x_k).$$

For linear features $V_w(x) = \phi(x)^\top w$:

$$w \leftarrow w + \alpha_k \delta_k \phi(x_k).$$

Section 9.4: Semi-Gradient TD

The book first writes the linear TD update as

$$w \leftarrow w + \alpha_k \underbrace{\left(r_k + \gamma \phi(x_k')^\top w - \phi(x_k)^\top w\right)}_{\text{TD error}} \phi(x_k).$$

Then introduce a separate vector $\widetilde{w}$ in the target:

$$w \leftarrow w + \alpha_k \left(r_k + \gamma \phi(x_k')^\top \widetilde{w} - \phi(x_k)^\top w\right) \phi(x_k).$$

With fixed $\widetilde{w}$, this is SGD on the sample loss and its solution satisfies

$$\Phi w^* = \Pi_{\phi,\xi} \mathcal{T}^\pi V_{\widetilde{w}}.$$

Book's fixed-point statement

At the fixed point $\widehat{V}^\pi = \Phi \widehat{w}$ of the projected operator,

$$\mathbb{E}_\pi \left[\left(R + \gamma \phi(X')^\top \widehat{w} - \phi(X)^\top \widehat{w}\right) \phi(X)\right] = 0.$$

From Value function approximation to Distributional Approximation

Want to linearly approximate probability distributions. However in general

$$\eta_{w_1}(x) + \eta_{w_2}(x) \notin \mathcal{P}(\mathbb{R})$$

and since the distribution lives in $\mathcal{P}(\mathbb{R})$, we can have non finite support.

How to solve this?

Fix a finite dimensional family of distributions $\mathcal{F}_\theta \subset \mathcal{P}(\mathbb{R})$ and then linearly approximate the parameters of the distribution. "Linear" distributional approximation means that the *parameters of a probability representation* depend linearly on features ϕ.

There are therefore two approximation layers:

1. a finite probability representation (categorical or quantile),
2. linear value function approximation.

Finite Return-Distribution Families

For a probability measure $\eta \in \mathcal{P}(\mathbb{R})$, we have

$$F_\eta(t) = \eta((-\infty, t]), \qquad F_\eta^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\eta(z) \geq \tau\}.$$

We consider the finite dimensional approximations:

$$\text{categorical:} \qquad \eta(x) = \sum_{i=1}^{m} p_i(x)\delta_{\theta_i}, \qquad p_i(x) \geq 0, \quad \sum_{i=1}^{m} p_i(x) = 1,$$

$$\text{quantile:} \qquad \eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}, \qquad \tau_i = \frac{2i-1}{2m}, \qquad \theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i).$$

What is state-dependent?

Categorical representations fix θ_i and learn the probabilities $p_i(x)$. Quantile fixes equal probabilities $1/m$ and learns locations $\theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i)$.

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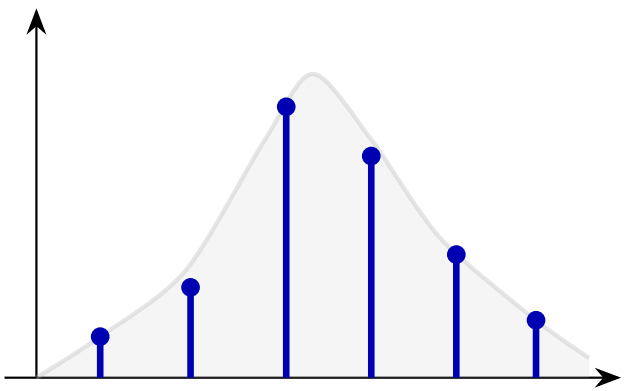
Question: We do not know $F_{\eta(x)}^{-1}$ in practice.

Categorical vs. Quantile

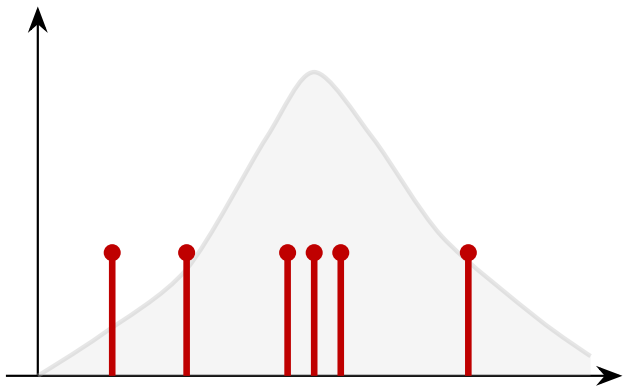
Categorical

$$\eta(x) = \sum_{i=1}^{m} p_i(x)\delta_{\theta_i}$$



Quantile

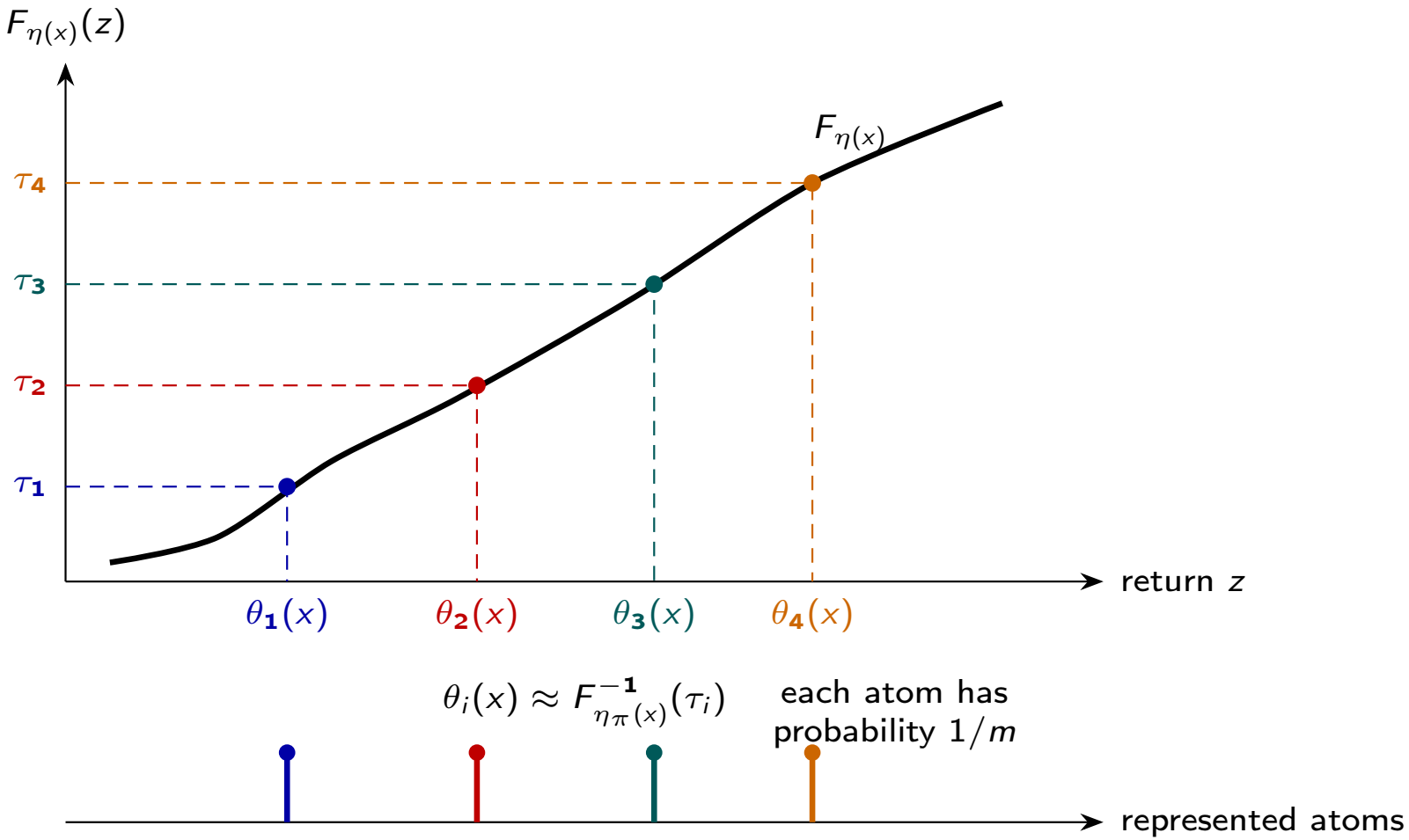
$$\eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}$$



Key distinction

τ_i is a fixed probability level. $\theta_i(x)$ is the return value placed at that level.

Quantile Levels: Input τ, Output θ



Changing τ_i asks for a different probability level; changing $\theta_i(x)$ moves the atom and therefore changes the represented distribution.

Distributional TD: Learning

For fixed feature map $\phi(x) \in \mathbb{R}^d$. A linear method has, for every atom,

$$u_i(x) = \phi(x)^\top w_i, \qquad w_i \in \mathbb{R}^d.$$

Observing a transition (x, r, x') gives us a empirical target. We fit this via

$$w_i \leftarrow w_i + \alpha\, \varepsilon_i(x, r, x')\, \phi(x).$$

where ε_i is a error term. This term and the meaning of u_i is what changes.

Linear Quantile TD: Representation and Loss

For the quantile representation, we choose

$$u_i(x) = \theta_i(x) = \phi(x)^\top w_i,$$

The represented return distribution is

$$\eta_w(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)} = \frac{1}{m}\sum_{i=1}^{m} \delta_{\phi(x)^\top w_i},$$

Then we take the quantile loss as in the tabular case

$$\rho_\tau(\Delta) = \left|\mathbb{1}_{\{\Delta<0\}} - \tau\right| |\Delta|.$$

For one target z, atom i uses this loss on the residual

$$L_{\tau_i}(w_i) = \rho_{\tau_i}\left(z - \phi(x)^\top w_i\right),$$

whose subgradient w.r.t. w_i is $-(\tau_i - \mathbb{1}_{\{z<\phi(x)^\top w_i\}})\phi(x)$.

Linear QTD: Distributional Semi-Gradient

Similar to the tabular case, for transition (x, a, r, x') we get m target samples

$$g_j = r + \gamma \phi(x')^\top w_j, \qquad j = 1, \dots, m.$$

For atom i, average the quantile loss over these targets:

$$L_i(w_i) = \frac{1}{m} \sum_{j=1}^{m} \rho_{\tau_i} \left(g_j - \phi(x)^\top w_i \right).$$

The semi-gradient step is

$$w_i \leftarrow w_i - \alpha \nabla_{w_i} L_i(w_i) = w_i + \alpha \frac{1}{m} \sum_{j=1}^{m} \underbrace{\left(\tau_i - \mathbb{1}_{\{g_j < \phi(x)^\top w_i\}} \right)}_{\text{quantile TD error}} \phi(x).$$

Unlike the scalar TD error, each averaged quantile error is in $[-1, 1]$.

Linear Categorical TD: Prediction

We approximate the probabilites $p_i(x; w)$ by a linear combination of features:

$$p_i(x; w) = \frac{\exp(\phi(x)^\top w_i)}{\sum_{j=1}^m \exp(\phi(x)^\top w_j)}, \qquad \eta_w(x) = \sum_{i=1}^m p_i(x; w)\delta_{\theta_i}.$$

transformed into probabilities by the softmax function. Softmax is needed, direct linear coefficients need not be nonnegative or sum to one.

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This can be handled, later!

Linear CTD: How to Stay on Fixed Support

For a sampled transition (x, r, x'), the Bellman update is an update of the prediction at the current state x:

$$\underbrace{\eta_w(x) = \sum_{j=1}^{m} p_j(x; w)\delta_{\theta_j}}_{\text{prediction at } x} \xrightarrow{\text{Observed } (r,x')} \underbrace{(b_{r,\gamma})_{\#}\eta_w(x') = \sum_{j=1}^{m} p_j(x'; w)\delta_{r+\gamma\theta_j}}_{\text{Empirical Bellman step for } x}.$$

Since in general $r + \gamma\theta_j$ need not lie on the fixed support, use the categorical projection Π_c to project the shifted atoms back to the fixed support $S = \{\theta_1, \ldots, \theta_m\}$. For each j, if $\theta_\ell \le r + \gamma\theta_j \le \theta_{\ell+1}$, split its mass as

$$\bar{p}_\ell \mathrel{+}= p_j(x'; w)\frac{\theta_{\ell+1} - (r + \gamma\theta_j)}{\theta_{\ell+1} - \theta_\ell}, \qquad \bar{p}_{\ell+1} \mathrel{+}= p_j(x'; w)\frac{(r + \gamma\theta_j) - \theta_\ell}{\theta_{\ell+1} - \theta_\ell}.$$

This gives the projected target

$$\bar{\eta}_{x,r,x'} = \Pi_c(b_{r,\gamma})_{\#}\eta_w(x') = \sum_{i=1}^{m} \bar{p}_i\delta_{\theta_i}.$$

Linear CTD: Target, Loss, and Update

To learn softmax probabilities, we use the Cross-entropy loss:

$$L(w) = -\sum_{i=1}^{m} \bar{p}_i \log p_i(x; w).$$

Since

$$\frac{\partial L}{\partial u_i} = p_i(x; w) - \bar{p}_i, \qquad \nabla_{w_i} u_i(x) = \phi(x),$$

gradient descent gives the semi-gradient CTD update:

$$w_i \leftarrow w_i + \alpha \underbrace{(\bar{p}_i - p_i(x; w))}_{\text{CTD error}} \phi(x).$$

Chapter 9 Objects: Signed Categorical Family

We can make our lives to develop theory easier by allowing negative measures. The signed categorical family on $S = \{\theta_1, \ldots, \theta_m\}$ is

$$\mathcal{F}_{S,m} = \left\{ \sum_{i=1}^{m} p_i \delta_{\theta_i} : p_i \in \mathbb{R}, \sum_{i=1}^{m} p_i = 1 \right\} \subseteq \mathcal{M}(\mathbb{R}).$$

Using this we can define the signed categorical return functions

$$\mathcal{F}_{S,m}^{\mathcal{X}} = \{\eta : \mathcal{X} \to \mathcal{F}_{S,m}\}.$$

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How to linearly approximate this?

Chapter 9 Objects: Linear Signed Approximation

Linear approximation, subtracting the mean!

$$p_i(x) = \frac{1}{m} + \phi(x)^\top w_i - \frac{1}{m}\sum_{j=1}^{m} \phi(x)^\top w_j.$$

Then

$$\sum_{i=1}^{m} p_i(x) = 1, \qquad p_i(x) \in \mathbb{R}.$$

The resulting family of distribution is denoted by $\mathcal{F}_{\phi,S,m}$.

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The resulting family of distribution is denoted by $\mathcal{F}_{\phi,S,m}$.

We want to perform bellman updates and ensure that the result is still in $\mathcal{F}_{\phi,S,m}$!

Chapter 9 Objects: Why Two Projections?

Start with a $\eta \in \mathcal{F}_{\phi,S,m}$. After the Bellman update,

$$\mathcal{T}^{\pi}\eta$$

two things can go wrong.

1. **Distribution representation.** For a state x, $(\mathcal{T}^{\pi}\eta)(x)$ need not be supported on $S = \{\theta_1, \ldots, \theta_m\}$. Hence we apply the categorical projection Π_c from before.
2. **Linear feature representation.** Even after Π_c, the map $x \mapsto (\Pi_c \mathcal{T}^{\pi}\eta)(x)$ need not lie in $\mathcal{F}_{\phi,S,m}$. Hence we need to define the feature projection by

$$\Pi_{\phi,\xi,d}\widetilde{\eta} \in \arg\min_{\eta' \in \mathcal{F}_{\phi,S,m}} d(\widetilde{\eta}, \eta').$$

In some suitable metric d. The final bellman update is then

$$\Pi_{\phi,\xi,d}\Pi_c \mathcal{T}^{\pi}\eta.$$

Chapter 9 Objects: Cramér Geometry

We use the Cramér distance introduced prior, given by

$$\ell_2^2(\nu, \nu') = \int_{\mathbb{R}} \left(F_\nu(z) - F_{\nu'}(z)\right)^2 \, dz$$

whenever this quantity is finite. Define

$$\mathcal{M}_{\ell_2,1}(\mathbb{R}) = \{\nu \in \mathcal{M}(\mathbb{R}) : \nu(\mathbb{R}) = 1, \ \ell_2(\nu, \delta_0) < \infty\}.$$

For return-distribution functions $\eta : \mathcal{X} \to \mathcal{M}_{\ell_2,1}(\mathbb{R})$ we can again define a weighted distance

$$\ell_{\xi,2}^2(\eta, \eta') = \sum_{x \in \mathcal{X}} \xi(x) \ell_2^2(\eta(x), \eta'(x)).$$

Chapter 9 Lemmas: Bellman and Categorical Steps

Assume rewards have finite first moments and the Markov chain under π has a unique stationary distribution ξ_π.

Lemma 9.14: Bellman Step

For any $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\ell_{\xi_\pi,2}(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \leq \gamma^{1/2} \ell_{\xi_\pi,2}(\eta, \eta').$$

Lemma 9.15: Categorical Projection

For any signed distributions ν, ν',

$$\ell_2(\Pi_c \nu, \Pi_c \nu') \leq \ell_2(\nu, \nu').$$

Chapter 9 Lemmas: Feature Projection

The final projection is the analogue of the scalar linear projection, but applied to signed categorical distributions with the Cramér metric.

Lemma 9.16: Linear Feature Projection

For any $\eta, \eta' \in \mathcal{F}^{\mathcal{X}}_{S,m}$,

$$\ell_{\xi_\pi,2}\left(\Pi_{\phi,\xi_\pi,\ell_2}\eta, \Pi_{\phi,\xi_\pi,\ell_2}\eta'\right) \leq \ell_{\xi_\pi,2}(\eta,\eta').$$

Thus each piece is either a contraction or a nonexpansion:

$$\mathcal{T}^\pi \quad \text{contracts,} \qquad \Pi_c,\ \Pi_{\psi,\xi_\pi,\ell_2} \quad \text{do not expand.}$$

Chapter 9 Theorem: Doubly Projected Bellman

Composing the three preceding statements gives the actual convergence-style result for linear distributional approximation in the book.

Theorem 9.13

Under the assumptions above, for all $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\ell_{\xi_\pi,2}\left(\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta, \Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta'\right) \leq \gamma^{1/2}\ell_{\xi_\pi,2}(\eta,\eta').$$

Hence the projected distributional Bellman operator has a unique fixed point in $\mathcal{F}_{\psi,S,m}$, by Banach's fixed-point theorem.

Chapter 9: Exact Contraction Chain

For $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\begin{aligned}
&\ell_{\xi_\pi,2}\left(\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta, \Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta'\right) \\
&\quad\leq \ell_{\xi_\pi,2}(\Pi_c\mathcal{T}^\pi\eta, \Pi_c\mathcal{T}^\pi\eta') \\
&\quad\leq \ell_{\xi_\pi,2}(\mathcal{T}^\pi\eta, \mathcal{T}^\pi\eta') \\
&\quad\leq \gamma^{1/2}\ell_{\xi_\pi,2}(\eta, \eta').
\end{aligned}$$

The three lines are exactly: feature projection nonexpansion, categorical projection nonexpansion, and distributional Bellman contraction.

From Fixed Features to Learned Features

A deep network replaces the hand-designed map $\phi(x)$ by a learned representation. Its penultimate layer can still be read as

$$\phi_w(x) \in \mathbb{R}^n,$$

followed by linear prediction heads.

- Earlier layers learn which state distinctions matter.
- Final layers map those features to values or distribution parameters.
- Backpropagation updates both simultaneously.

Linear approximation is therefore not discarded; it remains the final layer and the conceptual model for the deep semi-gradient update.

From DQN to Distributional DQN

Standard DQN predicts one scalar per action:

$$x \longmapsto \big(Q_w(x,a)\big)_{a\in\mathcal{A}}, \qquad Q_w(x,a) \approx \mathbb{E}[Z(x,a)].$$

Distributional DQN predicts a return distribution per action:

$$x \longmapsto \big(\eta_w(x,a)\big)_{a\in\mathcal{A}}, \qquad \eta_w(x,a) \approx \mathcal{D}(Z(x,a)).$$

For a transition (x,a,r,x'), the target is now a shifted future-return distribution:

$$(b_{r,\gamma})_{\#}\eta_{\widetilde{w}}(x',a^*), \qquad a^* \in \arg\max_{a'\in\mathcal{A}} Q_{\widetilde{w}}(x',a').$$

The output has m distribution parameters per action:

$$x \longmapsto \mathbb{R}^{|\mathcal{A}|\times m}.$$

- **C51:** using the categorical representation
- **QR-DQN:** using the quantile representation
- **IQN:** quantile level τ becomes an additional input.

C51: Prediction and Behavior

Using state action categorical representation with fixed $\theta_1 < \cdots < \theta_m$, the network outputs $\phi_i(x, a; w)$ and

$$p_i(x, a; w) = \frac{e^{\phi_i(x,a;w)}}{\sum_{j=1}^{m} e^{\phi_j(x,a;w)}}, \qquad \eta_w(x, a) = \sum_{i=1}^{m} p_i(x, a; w)\delta_{\theta_i}.$$

The induced action value is

$$Q_w(x, a) = \sum_{i=1}^{m} p_i(x, a; w)\theta_i.$$

Original C51 uses $m = 51$ evenly spaced atoms.

C51 Learning Algorithm

Select the target-network greedy action

$$a_{\widetilde{w}}(x') \in \arg\max_{a'} Q_{\widetilde{w}}(x', a').$$

Construct and project the target

$$\bar{\eta}(x, a) = \Pi_c(b_{r,\gamma})_{\#}\eta_{\widetilde{w}}(x', a_{\widetilde{w}}(x')) = \sum_{j=1}^{m} \bar{p}_j \delta_{\theta_j}.$$

Then minimize the cross entropy loss $L(w) = -\sum_{j=1}^{m} \bar{p}_j \log p_j(x, a; w)$ and update

$$w \leftarrow w + \alpha \sum_{i=1}^{m} (\bar{p}_i - p_i(x; w)) \nabla_w \theta_i(x, y; w).$$

This is pretty much the same as the linear CTD update, aside from the need to update vector valued.

QR-DQN: Quantile Prediction

QR-DQN interprets the m outputs as locations of an equally weighted quantile distribution:

$$\eta_w(x, a) = \frac{1}{m} \sum_{i=1}^{m} \delta_{\theta_i(x,a;w)}, \qquad Q_w(x, a) = \frac{1}{m} \sum_{i=1}^{m} \theta_i(x, a; w).$$

With $a_{\widetilde{w}}(x')$ greedy by this mean, the pairwise residuals are

$$\Delta_{ij} = r + \gamma \theta_j(x', a_{\widetilde{w}}(x'); \widetilde{w}) - \theta_i(x, a; w).$$

Every predicted quantile is compared with every target quantile.

QR-DQN: Quantile Huber Loss

For $\tau_i = (2i-1)/(2m)$, QR-DQN minimizes

$$L(w) = \frac{1}{m}\sum_{i,j=1}^{m} \rho_{\tau_i}^H(\Delta_{ij}), \qquad \rho_\tau^H(u) = |\mathbb{1}_{\{u<0\}} - \tau| H_1(u),$$

where

$$H_1(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le 1, \\ |u| - \frac{1}{2}, & |u| > 1. \end{cases}$$

Difference to linear case

- We use the Huber loss to reduce sensitivity to outliers and have a differentiable loss.
- The loss is over all weights simultaneously, not entrywise.

Particle Distributional DQN

QR-DQN represents

$$\eta_w(x,a) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x,a;w)}$$

where θ_i are quantile levels. MMD-DQN and SinkhornDRL keep the same empirical-measure form

$$\eta_w(x,a) = \frac{1}{m}\sum_{i=1}^{m} \delta_{z_i(x,a;w)},$$

and treat the z_i as unrestricted deterministic samples.

Overview of Particle Distributional DQN

1. Define a divergence d between return distributions.
2. Study whether $\mathcal{T}^\pi$ contracts under $\bar{d}(\eta,\nu) = \sup_{x,a} d(\eta(x,a),\nu(x,a))$.
3. Use the empirical version of d as a differentiable DQN loss.

This is a different route from deriving a quantile regression loss.

MMD-DQN: Maximum Mean Discrepancy

Let $\mathcal{H}_k$ be the RKHS of a continuous kernel k and let

$$\psi_\mu = \int k(z, \cdot)\, d\mu(z) \in \mathcal{H}_k$$

be the kernel mean embedding. The maximum mean discrepancy is

$$\mathrm{MMD}_k(\mu, \nu) = \|\psi_\mu - \psi_\nu\|_{\mathcal{H}_k}.$$

Equivalently, if $Z, Z' \sim \mu$ and $Y, Y' \sim \nu$ independently,

$$\mathrm{MMD}_k^2(\mu, \nu) = \mathbb{E}k(Z, Z') + \mathbb{E}k(Y, Y') - 2\mathbb{E}k(Z, Y).$$

- If k is characteristic, MMD is a metric on probability measures.
- Gaussian kernels are characteristic, so they can distinguish distributions.
- The metric geometry depends completely on the kernel.

MMD-DQN: Supremum Metric and Contraction

Nguyen et al. define the distributional-RL metric

$$\mathrm{MMD}_{\infty,k}(\eta,\nu) = \sup_{x,a} \mathrm{MMD}_k(\eta(x,a),\nu(x,a)).$$

Their contraction result requires more than "k is characteristic." A kernel must be:

$$k(x+c,y+c) = k(x,y) \quad \text{and} \quad \mathrm{MMD}_k((b_{\gamma,0})_{\#}\mu,(b_{\gamma,0})_{\#}\nu) = \gamma^{\alpha/2}\,\mathrm{MMD}_k(\mu,\nu).$$

Here $b_{\gamma,0}(z) = \gamma z$ and α is the scale-sensitivity order.

Theorem idea

For mixtures of shift-invariant, scale-sensitive kernels,

$$\mathrm{MMD}_{\infty,k}(\mathcal{T}^{\pi}\eta,\mathcal{T}^{\pi}\nu) \le \gamma^{\alpha/2}\,\mathrm{MMD}_{\infty,k}(\eta,\nu).$$

So $\mathcal{T}^{\pi}$ has the same fixed-point story as in Wasserstein, but only under suitable MMD kernels.

MMD-DQN: Important Caveat

The paper separates theory and best empirical practice:

- Unrectified kernels such as

$$k_\alpha(x, y) = -|x - y|^\alpha, \qquad 0 < \alpha < 2,$$

  give the clean contraction theorem.
- Gaussian kernels are characteristic and useful for moment matching.
- But the paper proves that $\mathcal{T}^\pi$ is *not* a contraction in MMD for Gaussian kernels in general.

Interpretation

MMD-DQN uses Gaussian-kernel MMD because it gives a practical particle loss: it attracts particles to Bellman target samples and repels particles from collapsing. The contraction theorem motivates the metric framework, but does not literally certify the Gaussian version used in Atari experiments.

MMD-DQN: Learning with Particles

The network outputs particles

$$Z_w(x,a) = \{z_i(x,a;w)\}_{i=1}^m, \qquad \eta_w(x,a) = \frac{1}{m}\sum_i \delta_{z_i(x,a;w)}.$$

For a transition (x,a,r,x'), construct Bellman target particles

$$y_j = r + \gamma z_j(x', a^*; \widetilde{w}), \qquad a^* \in \arg\max_{a'} \frac{1}{m}\sum_j z_j(x', a'; \widetilde{w}).$$

The biased empirical MMD loss is

$$L_{\mathrm{MMD}}(w) = \frac{1}{m^2}\sum_{i,i'} k(z_i, z_{i'}) + \frac{1}{m^2}\sum_{j,j'} k(y_j, y_{j'}) - \frac{2}{m^2}\sum_{i,j} k(z_i, y_j).$$

Thus the metric is not only an evaluation tool: it becomes the backpropagated Bellman error between two empirical return distributions.

SinkhornDRL: Regularized Wasserstein Loss

Sun et al. start from optimal transport. For cost $c(z, y)$,

$$\mathrm{OT}_c(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int c(z, y)\, d\pi(z, y).$$

Entropic regularization gives

$$\mathrm{OT}_c^{\varepsilon}(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int c\, d\pi + \varepsilon\, \mathrm{KL}(\pi \mid \mu \otimes \nu).$$

The debiased Sinkhorn divergence is

$$S_c^{\varepsilon}(\mu, \nu) = \mathrm{OT}_c^{\varepsilon}(\mu, \nu) - \frac{1}{2}\,\mathrm{OT}_c^{\varepsilon}(\mu, \mu) - \frac{1}{2}\,\mathrm{OT}_c^{\varepsilon}(\nu, \nu).$$

It is a regularized Wasserstein-type loss: smoother than exact OT and computable by Sinkhorn iterations.

SinkhornDRL: Contraction Result

Define the supremal Sinkhorn divergence

$$\bar{S}_c^{\varepsilon}(\eta, \nu) = \sup_{x,a} S_c^{\varepsilon}(\eta(x,a), \nu(x,a)).$$

The paper analyzes contraction through two properties:

- **Sum invariance:** adding the same independent reward noise does not increase the distance.
- **Scale sensitivity:** discounting returns shrinks the distance.

For the cost family used in their theorem, the distributional Bellman operator satisfies a bound of the form

$$\bar{S}_c^{\varepsilon}(\mathcal{T}^{\pi}\eta, \mathcal{T}^{\pi}\nu) \leq \rho_{\varepsilon,\gamma} \bar{S}_c^{\varepsilon}(\eta, \nu), \qquad \rho_{\varepsilon,\gamma} < 1.$$

The constant depends on γ, the regularization ε, the cost order, and the return-distribution set of the finite MDP.

SinkhornDRL: Interpolating Wasserstein and MMD

The paper emphasizes that Sinkhorn divergence interpolates between two familiar geometries:

$$\varepsilon \to 0 \quad \Rightarrow \quad S_c^\varepsilon \text{ behaves like Wasserstein/OT,}$$

$$\varepsilon \to \infty \quad \Rightarrow \quad S_c^\varepsilon \text{ behaves like an MMD-type distance.}$$

Correspondingly, their theorem recovers contraction behavior consistent with both endpoints, and for finite ε proves a genuine contraction with a factor between the Wasserstein contraction factor and 1.

Why this matters

Regularization makes the transport plan smoother and the loss easier to optimize, but it weakens the contraction factor compared with exact Wasserstein. The point is controlled trade-off, not free improvement.

SinkhornDRL: Learning with Particles

As in MMD-DQN, the return distribution is represented by particles:

$$\eta_w(x, a) = \frac{1}{m} \sum_i \delta_{z_i(x,a;w)}.$$

For a Bellman target particle set $\{y_j\}_{j=1}^m$, form the cost matrix

$$C_{ij} = c(z_i, y_j).$$

With uniform weights $a = b = (1/m, \ldots, 1/m)$, Sinkhorn iterations approximate

$$\min_{\Pi \in U(a,b)} \sum_{i,j} \Pi_{ij} C_{ij} + \varepsilon \sum_{i,j} \Pi_{ij}(\log \Pi_{ij} - 1),$$

followed by the self-cost debiasing terms in S_c^ε.

DQN update

Replace the QR-DQN quantile Huber loss by the empirical Sinkhorn divergence between current particles and Bellman target particles.

Metric-Based Particle DQN: Comparison

Method	Representation	Metric/loss	Theory
QR-DQN	quantiles	quantile Huber	W_1 projection
MMD-DQN	particles	MMD	conditional contraction
SinkhornDRL	particles	Sinkhorn divergence	contraction theorem

The common Bellman target is still

$$y_j = r + \gamma z_j(x', a^*; \widetilde{w}), \qquad a^* = \arg\max_{a'} \frac{1}{m} \sum_j z_j(x', a'; \widetilde{w}).$$

The new idea is that once a distribution metric has a Bellman contraction story, its empirical sample version can be used directly as a learning objective.

Takeaways

1. Linear approximation introduces state aliasing and a weighted projection problem; the sampling distribution is central to stability.
2. Semi-gradient TD tracks a projected Bellman process, and target weights make that relationship explicit.
3. Linear QTD and CTD make distribution *parameters* linear in features.
4. Deep methods learn the features and retain the same prediction–target–semi-gradient structure.
5. C51, QR-DQN, and IQN differ mainly in how they parameterize and train the output distribution; that choice also shapes the learned representation.
6. MMD-DQN and SinkhornDRL are metric-first particle methods: they define a distribution distance, analyze Bellman contraction, then use the empirical distance as the DQN loss.

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Chapter 10: Why This Is Not a DQN Theorem

C51, QR-DQN, and IQN combine several ingredients:

$$\text{nonlinear network} + \text{bootstrapping} + \text{greedy control} + \text{replay/target networks}.$$

The theory is therefore much weaker than for policy evaluation.

What Theorem 9.13 needs

A fixed policy, linear projection in a fixed metric, and a fixed operator

$$\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi.$$

What Chapter 10 uses

Nonlinear parameters and a moving control target, e.g.

$$r + \gamma Z_{\widetilde{w}}\left(x', \arg\max_{a'} \mathbb{E}[Z_{\widetilde{w}}(x', a')]\right).$$

So Chapter 10 inherits the projection intuition, but not a comparable general convergence theorem for deep distributional DQN.

IQN: Make the Quantile Level an Input

C51 and QR-DQN output a fixed finite parameter vector. IQN instead models the quantile function itself:

$$(x, a, \tau) \longmapsto \theta_w(x, a, \tau) \approx F^{-1}_{\eta(x,a)}(\tau).$$

The level is encoded by

$$\varphi(\tau) = (\cos(\pi \tau i))_{i=0}^{M-1},$$

transformed by a fully connected layer, and combined multiplicatively with the state features. The final action heads satisfy

$$\theta_w(x, a, \tau) = \phi_w(x, \tau)^\top w_a.$$

IQN: Behavior and Learning

Approximate the mean with independent $\tau_i \sim \mathcal{U}(0,1)$:

$$Q_w(x,a) \approx \frac{1}{m}\sum_{i=1}^{m} \theta_w(x,a,\tau_i).$$

For independent online levels τ_i and target levels τ'_j, minimize pairwise losses of

$$r + \gamma\theta_{\widetilde{w}}(x', a_{\widetilde{w}}(x'), \tau'_j) - \theta_w(x,a,\tau_i).$$

Sampling many level pairs reduces gradient variance. IQN approximates the whole quantile function without committing to fixed output locations.

IQN Makes Risk Sensitivity Simple

Risk-neutral behavior samples $\tau \sim \mathcal{U}(0,1)$. Lower-tail CVaR at level $\bar{\tau}$ is

$$\mathrm{CVaR}_{\bar{\tau}}(G_w(x,a)) = \frac{1}{\bar{\tau}} \int_0^{\bar{\tau}} \theta_w(x,a,\tau)\, d\tau.$$

Approximate it by changing only the sampling distribution:

$$\mathrm{CVaR}_{\bar{\tau}}(G_w(x,a)) \approx \frac{1}{m} \sum_{i=1}^{m} \theta_w(x,a,\tau_i), \quad \tau_i \sim \mathcal{U}(0,\bar{\tau}).$$

Thus an implicit representation separates learning the return distribution from choosing how that distribution should guide behavior.

How Distributional Predictions Shape Features

Split the network into learned features and final prediction heads.

- DQN trains the representation to predict one conditional mean per action.
- C51 trains it through m probability errors per action.
- QR-DQN/IQN train it through many pairwise quantile residuals.

The type and number of predictions therefore change the learned state representation. This richer auxiliary prediction problem is a leading explanation for improved control performance even when actions are ultimately selected by the mean.